MODEL PROCEDURE

EX. NO. 2.1 – Create a 3D Solid Model from Isometric Drawing

STEP-1:

Start the Sketch Command

- Select Sketch → Create Sketch.

STEP-2:

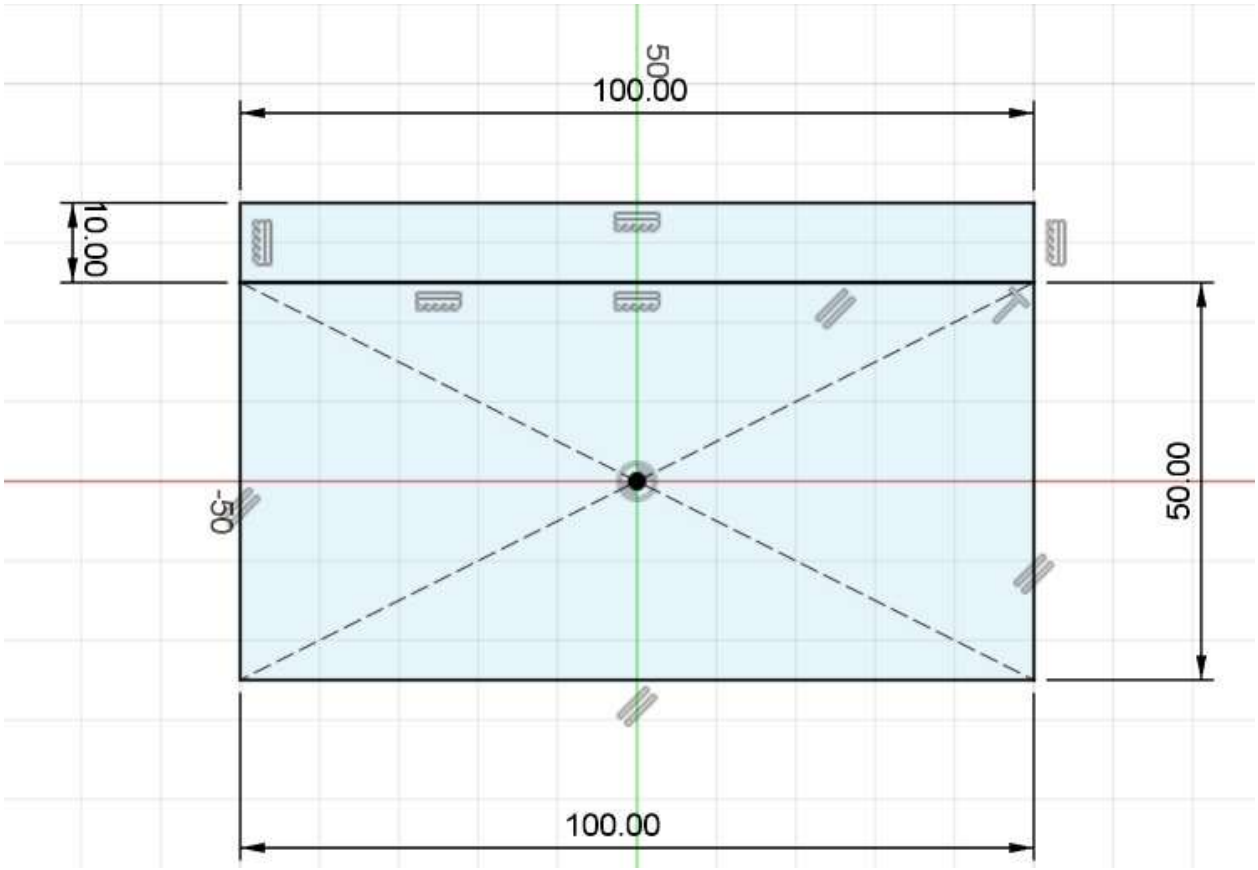
Select the Sketch Plane

- Select the Front (XZ) Plane.

STEP-3:

Create a Rectangle

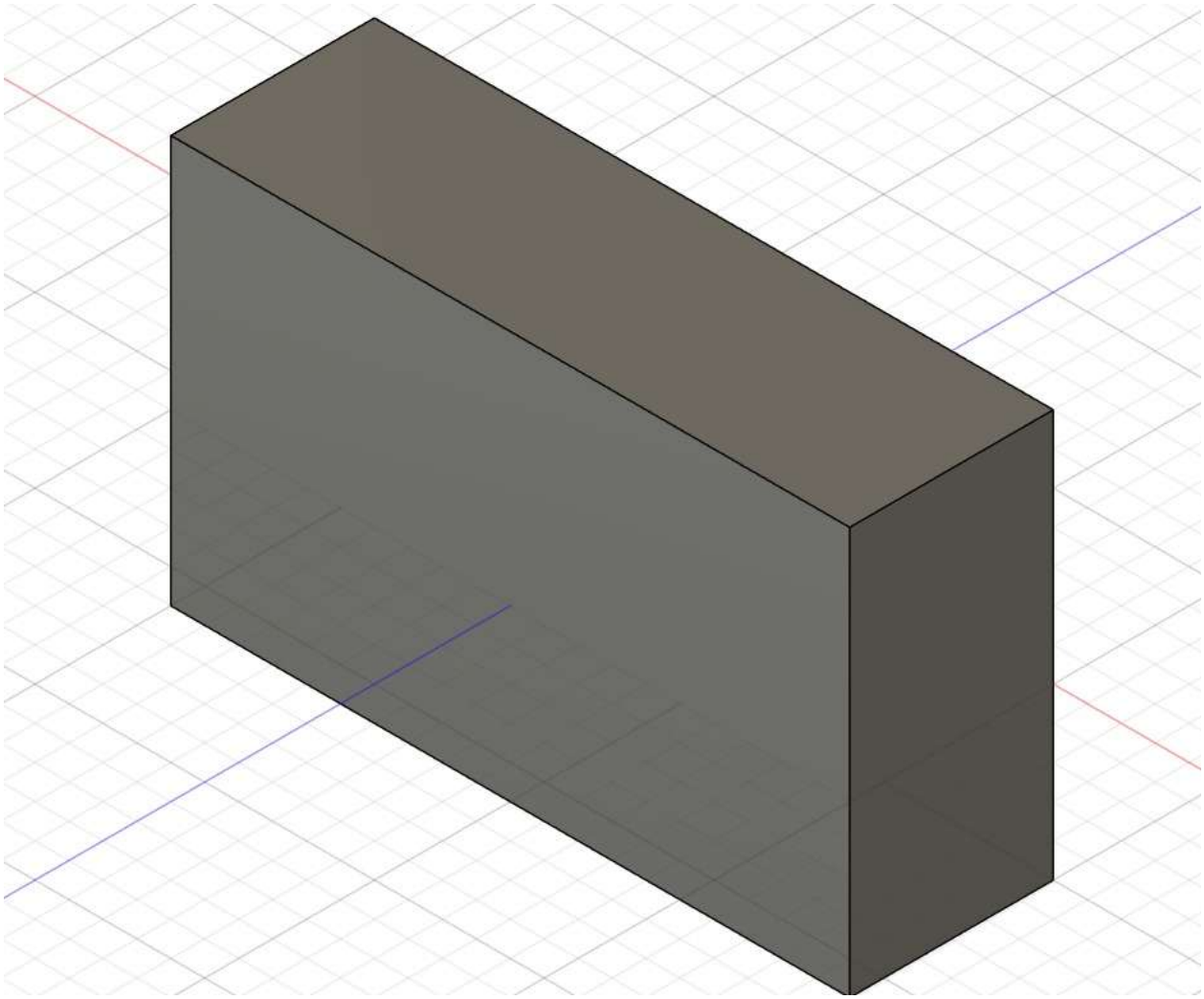
- **Select Sketch → Rectangle → Center Rectangle.**
- **Hover over the origin (center) of the sketch.**
- **Click once to begin placing the rectangle.**
- **Press the Tab key and enter the width as 100 mm and height as 60 mm.**
- **Click again to place the rectangle.**
- **Click Finish Sketch.**



STEP-4:

Extrude the Rectangle

- **Select Extrude.**
- **Select the rectangle sketch.**
- **Enter the extrusion distance as 30 mm.**
- **Press Enter to accept the value.**



STEP-5:

Start a New Sketch

- Select Sketch → Create Sketch.

STEP-6:

Select the Sketch Plane

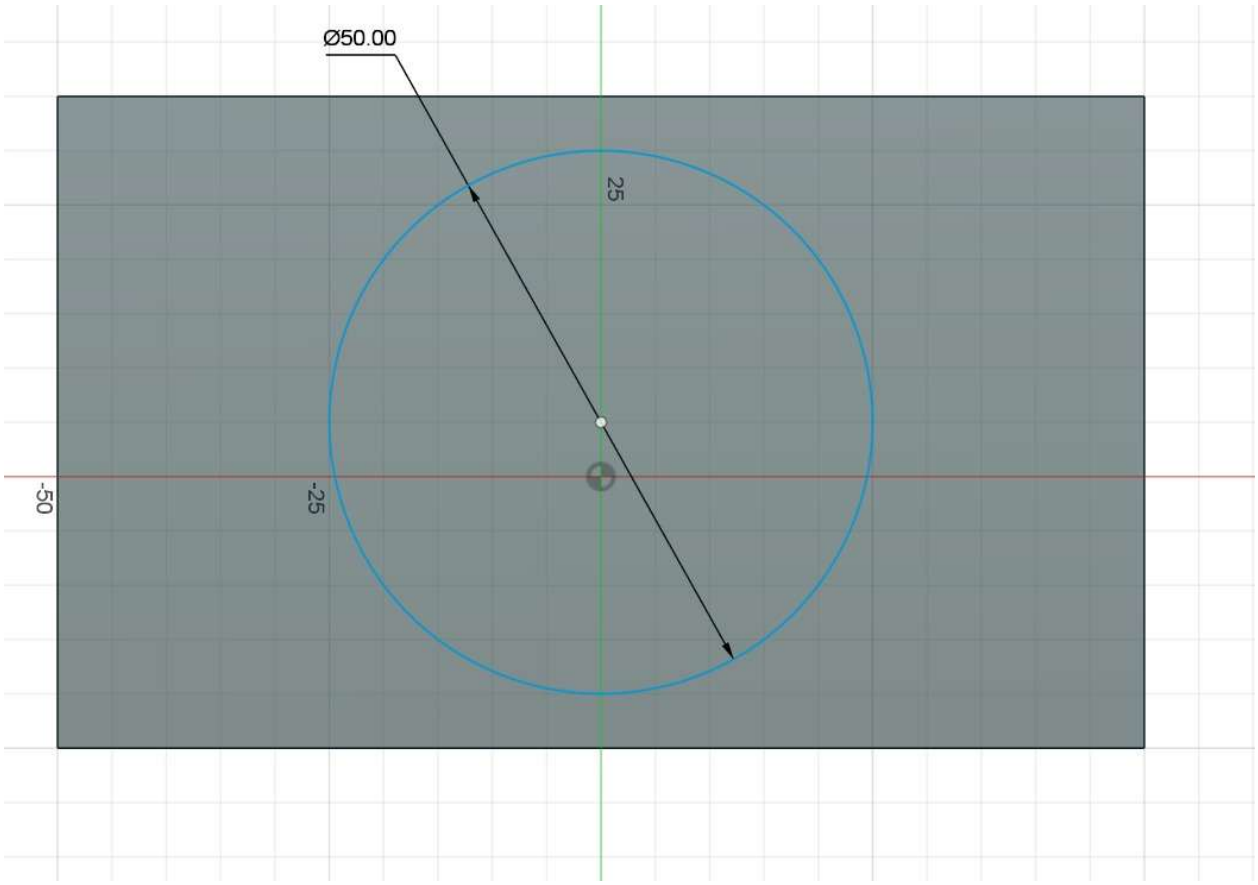
- Select the front surface of the rectangular block.

STEP-7:

Create a Circle

- **Select Sketch → Circle → Center Circle.**
- **Hover over the origin (center) of the sketch.**
- **Click once to begin placing the circle.**

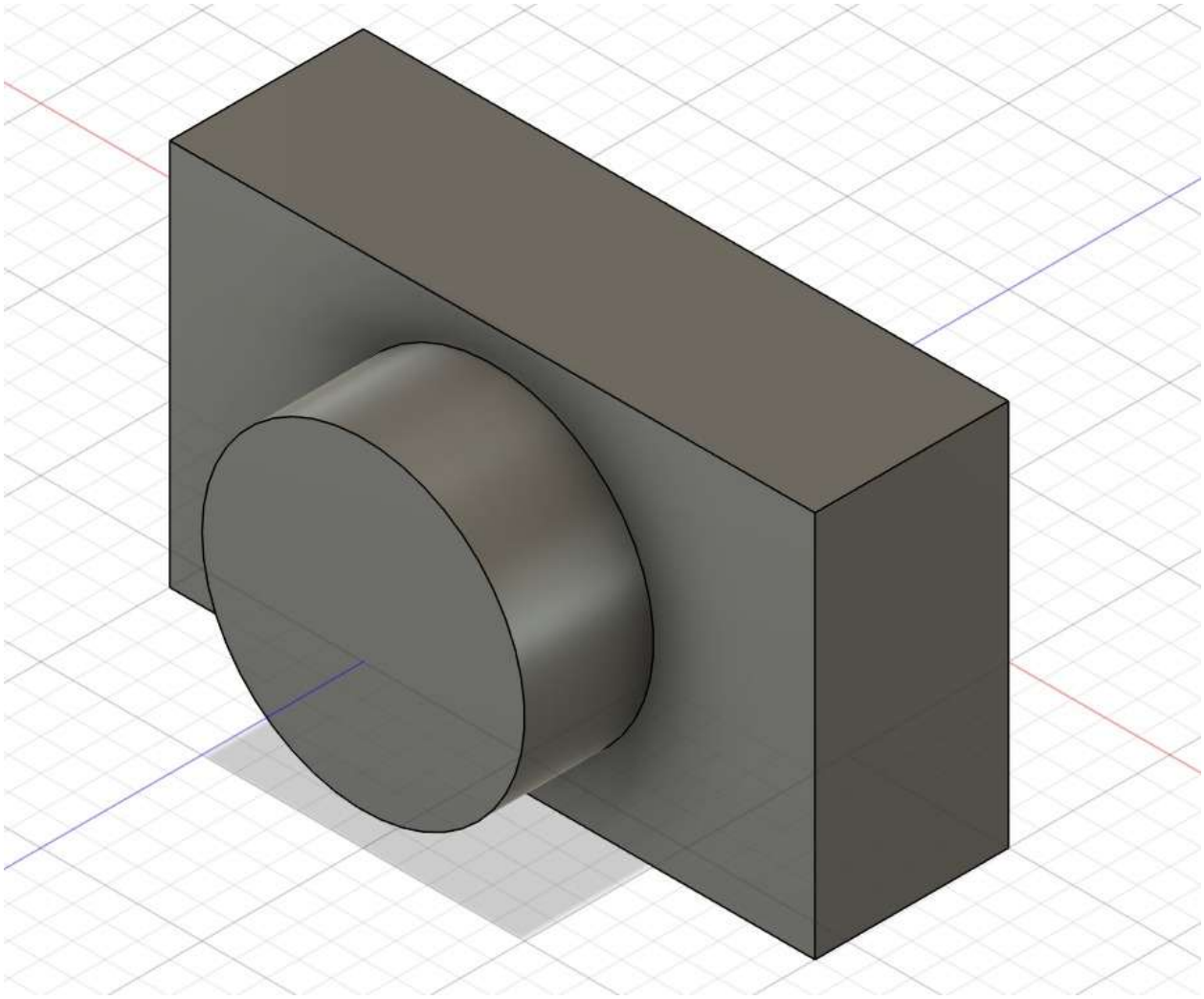
- **Press the Tab key and enter the diameter as 100 mm.**
- **Click again to place the circle.**
- **Click Finish Sketch.**



STEP-8:

Extrude the Circle

- **Select Extrude.**
- **Select the circle profile.**
- **Enter the extrusion distance as 20 mm.**
- **Press Enter to accept the value.**



STEP-9:

Start a New Sketch

- Select Sketch → Create Sketch.

STEP-10:

Select the Sketch Plane

- Select the top surface of the rectangle.

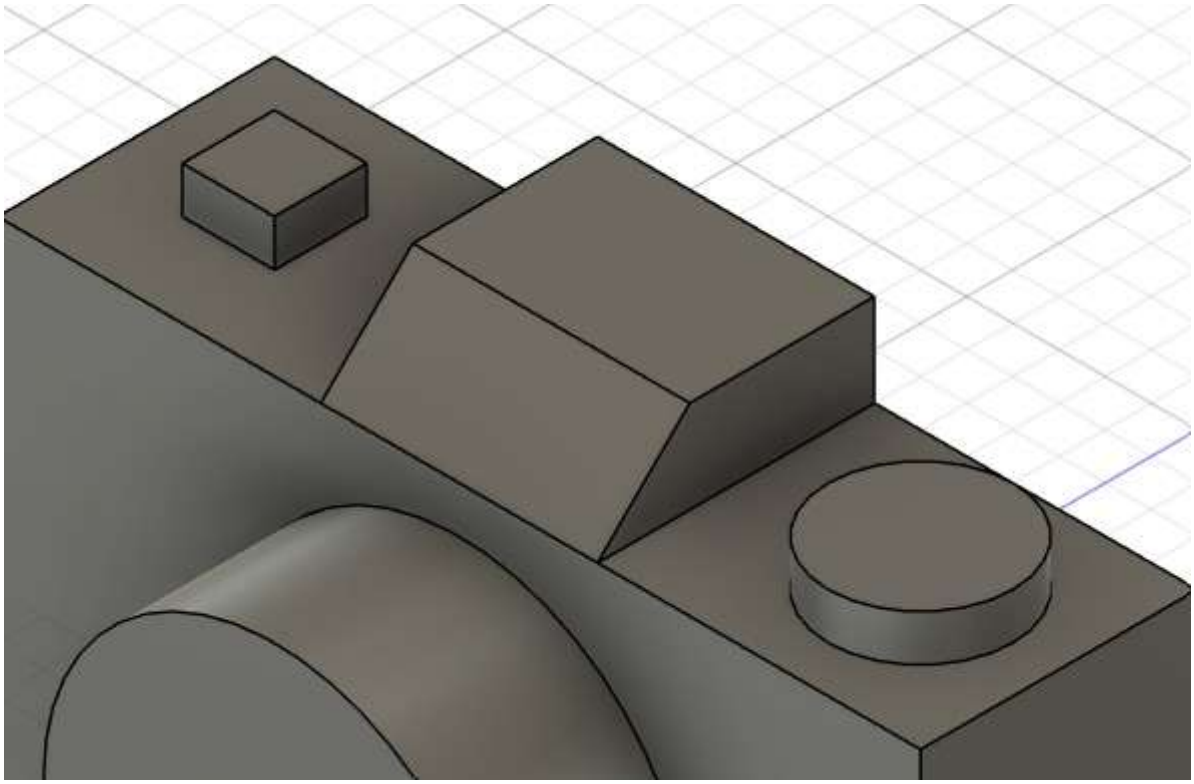
STEP-11:

Select the Line command and draw the required projections according to the given diagram. Click Finish Sketch, select Extrude, choose the required closed profile, and enter the extrusion distance specified in the drawing. Click OK.

Select the Line command and draw the required projections according to the given diagram. Click Finish Sketch, select Extrude, choose the required closed profile, and enter the extrusion distance specified in the drawing. Click OK.

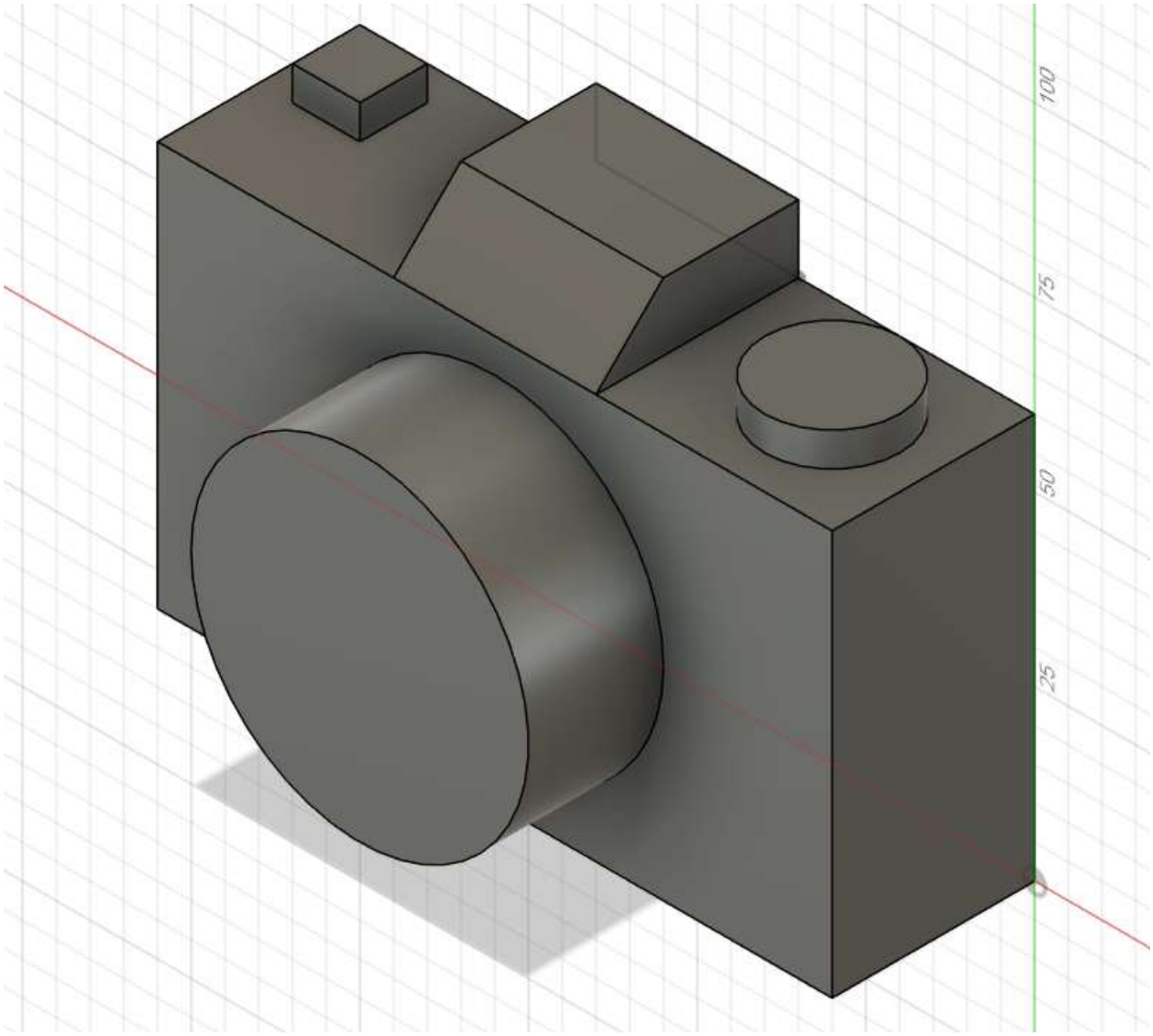
STEP-12: Apply Chamfer

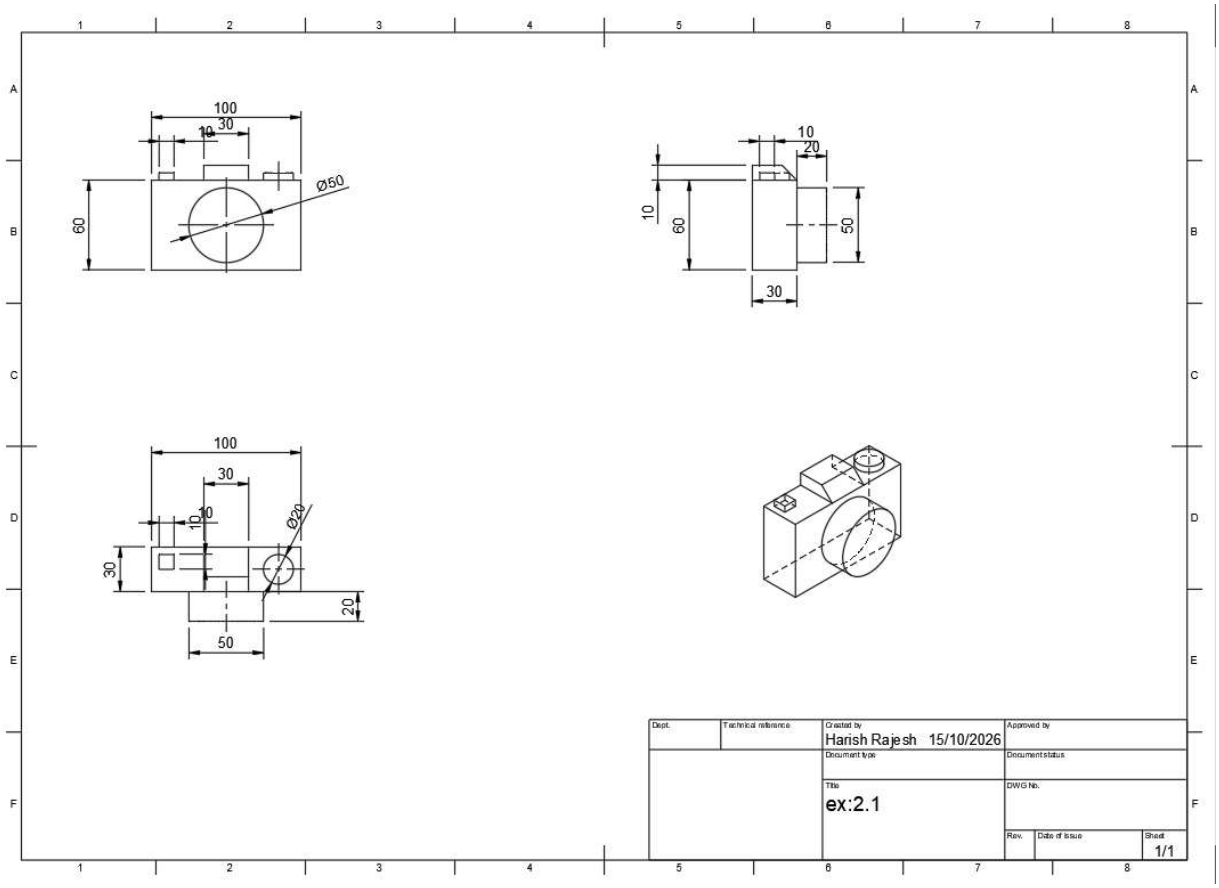
Select Solid → Modify → Chamfer. Select the required sharp edges of the 3D model, enter the chamfer distance, preview the result, and click OK to apply the chamfer.



STEP-13: Verify the Model

Rotate the model using the ViewCube to inspect it from different angles. Check that the model matches the given isometric drawing and save the completed design.


100
75
50
25



RESULT: Thus, the 3D solid model is created from the given isometric drawing using Autodesk Fusion 360, and the required chamfer is applied to the selected edges.